

Interpretable CNN Framework for Student Drug Detection Using Academic and Legal Indicators

Dip Kumar Saha, Tuhin Hossain, Rounakul Islam Boby, M. F. Mridha, *Senior member, IEEE.* , and and Zeyar Aung *Senior member, IEEE.*

Abstract—Drug misuse is a growing global health issue, particularly among university students, leading to self-harm and severe social consequences. Addicted students often exhibit distinct behavioral, social, and familial patterns, underscoring the need for reliable early detection. This study proposed an explainable ensemble framework (method-4) for detecting student drug addiction (SDA) which based on novel 1D CNN accompanied by a full linked system (FLSs) (method-1), and 1D CNN–LSTM–FLS (method-2). Both Method-1 and Method-2 were enhanced through the incorporation of a unit of feature layers (UFLs) to improve feature representation. The methods were trained on 12,745 student records from university and rehabilitation centers, incorporating academic, social, financial, health, and legal features. With integration of explainable artificial intelligence (XAI) via shapley additive explanations (SHAP), Method-4 achieved superior results: F1-score 69.55%, accuracy 74.00%, precision 83.92%, sensitivity 59.38%, specificity 88.62%, log loss 9.30%, and misclassification error 26.00%. The proposed XAI-enhanced ensemble further improved accuracy by 12–15% over the non-XAI approach, highlighting its role in enhancing model stability and interpretability. The results demonstrate that integrating XAI markedly improves the methods stability and ensures fairer detections.

Index Terms—Explainable AI, 1-D CNN, Stacking ensemble, Drug addiction

I. INTRODUCTION

Drug abuse is an issue that should concern every nation on Earth. As far as modern society’s economic and social issues are concerned, it ranks first [1]. When it comes to this issue, every nation has its own unique perspective and scale. The World Drug Report 2014 estimates that there will be 292 million drug users in 2022, a 20% increase from the previous decade. Substance abuse affects almost 64 million people, making drug use extremely risky and necessitating treatment [2]. Addiction to drugs is associated with norms of conduct in one’s family and community. It has a direct correlation to issues with public health and has a detrimental effect on people’s mental health. Among other things, using amphetamines like meth and amyl nitrite, sometimes called “poppers,” increases the risk of engaging in sexually risky behaviors and the likelihood of contracting HIV [3]–[5]. People of all genders who drink alcohol are more likely to act aggressively [6]. Once again, it has been found that

spending time with loved ones influences a person’s choice to start smoking [7]. This provides more evidence that narcotics have an impact on people’s social and behavioral issues. This provides more evidence that narcotics have an impact on people’s social and behavioral issues. Drugs ruin a person’s personal and professional lives in addition to their health. We therefore believed that by studying a person’s everyday social activities, including family dynamics, financial situation, health problems, etc., we might determine whether or not they are involved in drug misuse. It is challenging to identify problematic drug use in a professional context, particularly because many patients are reluctant or unable to completely disclose their patterns of drug use, whether prescribed or illegal. This is why a growing number of healthcare systems are relying on administrative and clinical databases to help find high-risk patients. These databases contain information about patients’ socioeconomic status, medical history, prescription drug use, substance use disorders, and health care utilization patterns. The particular statistical and analytic challenges posed by large datasets are a result of their extreme complexity and constant expansion.

Machine learning (ML), a rapidly expanding field of computational algorithms, aims to mimic human intelligence through environmental learning [8]. The capacity of ML-based modeling to handle complicated data and deliver personalized predictions has made it a major tool for analyzing social data at the individual level to estimate patient risk. ML’s capacity [9] to provide individual-level optimal predictions makes it a useful addition to conventional statistical analysis [10], [11]. In contrast to traditional statistical methods, which primarily focus on group-level significance in the current data, ML includes an inherent procedure to evaluate the predictive performance on fresh data (e.g., cross-validation [12]), guaranteeing that the method’s prediction may be applied to different domains. Instead of depending on a priori assumptions, ML methods are typically data-driven, meaning they can automatically and objectively optimize their parameters and predictive variables to capture the information and patterns inside high-dimensional data. Essential for practical uses involving individual outcomes, these methods can also generate forecasts at the individual level in addition to the group level [13].

A quantitative evaluation on 12,745 student records from universities and rehabilitation centers showed that the proposed XAI-enhanced ensemble framework (Method-4) achieved an F1-score of 69.55%, accuracy of 74.00%, precision of 83.92%, sensitivity of 59.38%, specificity of 88.62%, log loss of 9.30%, and misclassification error of 26.00%. This represents a 12–15% accuracy improvement over the non-XAI

Department of CSE, Stamford University Bangladesh, Dhaka, Bangladesh (e-mail: dip9178@gmail.com, rounakul.islam@stamforduniversity.edu.bd)
Department of CSE, University of Scholars, Dhaka, Bangladesh (e-mail: tuhin97.hossain@gmail.com).
Department of Computer Science, American International University-Bangladesh, Dhaka, Bangladesh (e-mail: firoz.mridha@aiub.edu).
Department of Computer Science, Khalifa University, Abu Dhabi, United Arab Emirates (e-mail: zeyar.aung@ku.ac.ae).

approach and baseline models, including SVM, LSTM, MLP, 1D CNN-FLS, and 1D CNN-LSTM-FLS. The framework exhibits strong robustness across diverse academic, social, financial, health, and legal features, while SHAP-based XAI enhances stability, interpretability, and fairness, providing a reliable solution for early-stage identification of SDA. Our key contributions are as follows-

- This study proposed an explainable ensemble framework for early-stage detection of SDA, integrating lightweight DL and hybrid models.
- Two hybrid models, 1D CNN-FLS and 1D CNN-LSTM-FLS, were developed and enhanced with UFLs for improved feature representation across academic, social, financial, health, and legal attributes, while SHAP-based XAI was incorporated to improve model interpretability, stability, and fairness.
- The UFL architecture, combining 1D convolution, 1D max-pooling, and dense layers, is lightweight, resource-efficient, and achieves reliable accuracy.
- The XAI-enhanced ensemble framework enables reliable and efficient early-stage detection of SDA, supporting timely intervention and informed decision-making in educational and rehabilitation settings.

II. LITERATURE REVIEW

Drug misuse is a critical global health concern, with over 292 million users reported in 2022. Accurate and interpretable detection, particularly among university students, can be achieved using CNN [14] and XAI [15] to address the increasing prevalence and complexity of substance abuse. Liu et al. [16] developed an ML model to predict opioid use disorder (OUD) using Canadian health data. The model was trained on data from 2014 to 2018 ($n = 699,164$) and validated using a hold-out sample from the same period ($n = 174,791$) and a separate 2019 sample ($n = 316,039$). OUD diagnoses were identified using ICD codes. Their model accurately predicted the occurrence of 6409 OUD patients in 2019 (mean [SD], 45.34 [14.28], 3400 men) with a high level of specificity 79%, sensitivity 93%, and balanced accuracy 86%. In another study, Saleem et al. [17] investigated the predictive capacity of the Theory of Planned Behavior (TPB) on illicit drug usage among Indian college students through a cross-sectional study involving 1,254 participants (47.1% male, 8.1% female, median age 22.61). The results of surveys were used to examine attitudes, perceived norms, self-efficacy, and intentions. According to structural equation modeling, TPB components accounted for 30% of the variation in drug use intentions. Also, Xie et al. [18] investigated OUD via social media posts to discover treatment impediments. They created the SINDEL DL model, which analyzes patient tales using word semantics and text similarity networks. SINDEL outperformed standard models and DL approaches. It outperformed BLSTM with 4.15% more precision, 11.96% more recall, and an 8.44% higher F1 score. Rekabdar et al. [19] tested ML and DL models to identify dangerous alcohol and drug users in a 6,978-hospitalized adult SBIRT program. It used Generative Adversarial Imputation Networks (GAIN) to manage miss-

ing data and Synthetic Minority Over-sampling Technique-Nominal Continuous (SMOTE-NC) to augment data. They used mixed-effects models on this processed data to predict AUDIT and DAST scores most accurately, proving GAIN imputation and combination modeling work. One more study, Hassan et al. [20] inspected 130 instances with 13 features from Bangladeshi colleges to determine the key risk factors for college student drug misuse. They analyzed the data using ML and identified essential aspects using association rule mining, notably the Apriori algorithm. The Apriori technique identified 8 significant rules with a confidence level of 0.90 and a support value of 0.45, indicating drug abuse risk factors. Alam et al. [21] investigated the reasons behind drug addiction and strategies for lowering its incidence among Bangladeshi men between the ages of 17 and 45. The study collected 188 qualitative data points from addiction recovery centers and non-addict educational institutions. Various ML algorithms, such as logistic regression (LR), decision tree (DT), random forest (RF), naive Bayes (NB), and support vector machine (SVM), were evaluated. For predicting addiction risk factors, the RF model fared best with 97.35% accuracy, followed by XGBoost at 96.28% and DT at 94.68%. Jing et al. [22] used ML to uncover environmental, psychological, and health aspects connected to substance use disorder (SUD) across age groups. In five assessment waves, data were gathered from 206 females and 494 males, as well as their parents, at the ages of 10, 12, 14, 16, 19, and 22. To identify the main predictors of SUD, more than a thousand variables were examined using the RF technique. The accuracy of the model increased to 86% by the age of 22, from 74% for those aged 10 to 12. In ages 10–22, RF outperformed other ML algorithms in detecting SUD risk. Mukhtar et al. [23] illustrated the application of CNN for classifying EEG signals to identify alcoholism, augmenting the model with kernel regularization, batch normalization, and dropout methodologies. These enhancements resulted in improved performance over time, and the study also informed researchers on the application of regularization in CNN design. The technique attained elevated precision, recall, and AUC, achieving accuracy levels of up to 98% on the UCI ML dataset. Ebrahimi et al. [24] used electronic health data to identify alcohol use problems using deep neural networks (DNN). To categorize people as normal, risky, or destructive drinkers, they developed a feedforward DNN with four fully connected layers, two ReLU, and a softmax activation. In comparison to conventional algorithms, the model was able to correctly identify approximately 96% of regular drinkers, 82% of risky drinkers, and 89% of harmful drinkers. Dong et al. [25] suggested a DL model based on LSTM to identify OUD in individuals with a history of opioid prescriptions. The model was trained using demographics, medical information, blood tests, prescriptions, and symptoms from five previous interactions. In comparison to DT, LR, RF, and dense neural networks, the LSTM demonstrated enhanced performance. It attained an F1 score of 0.8023 (0.016 more than DNN) and an AUROC of 0.9369 (0.145 higher), indicating robust predictive accuracy. Anggrawan et al. [26] developed a ML-based system to simulate human expertise to identify drug users and their medications. They used certainty factor and forward chaining

to build the ML-based expert system. The results reveal that the ML-based expert system can detect drug users and types with 80% accuracy. Marcon et al. [27] employed elastic net regularization, RF, and artificial neural networks(ANN) to predict medical students' high-risk drinking (HRD) based on personal information, mental health, university status, and sociodemographic. All three models obtained comparable outcomes, with average AUC scores of approximately 0.72 for both the training and testing sets. Their findings imply that ML models could detect students at risk for alcohol use disorder (AUD) using numerous indicators of excessive drinking at low cost and speed. Arif et al. [28] employed ML methods to forecast the risk of drug addiction, identifying significant factors through surveys, literature reviews, and conversations with addicts and physicians. After pre-processing data, nine common ML algorithms were applied and assessed using numerous metrics. Logistic regression surpassed other classifiers with 97.91% accuracy. In contrast, when principal component analysis was applied, CART's accuracy fell to 59.37%, demonstrating poor performance. Kinreich et al. [29] utilized COGA-rich datasets, which included electroencephalogram (EEG), genetic, and family history data from individuals as young as 12, to predict the development of AUD. They developed a ML model that incorporated gender and family history, along with multidimensional variables, to identify individuals at risk of AUD. Their findings indicated that prediction models were more accurate for African American (AA) samples compared to European American (EA) samples. Gong et al. [30] investigated how impulsivity, alexithymia, psychological capital, interpersonal confidence, life events, familial intimacy, aggressive conduct, parental conflict, and deviant peers affect substance appetite in 895 male users. While aggressive conduct was the primary psychological element, life events had the greatest influence on desire, according to gradient boosting analysis. The findings promote prevention and intervention by revealing the pros and cons of various influences on substance craving. Hassan et al. [31] investigated the factors influencing drug addiction risk among students through the analysis of 130 data instances from multiple universities in Bangladesh, employing ML methodologies. Using the Apriori technique and association rule mining, they found eight rules with a confidence level of 0.90 and a support of 0.45. The results indicated that the main causes of drug addiction were marital status, environmental variables, and personal struggles. Deng et al. [32] gathered drug-drug interaction (DDI) data from DrugBank and utilized natural language processing (NLP) to divide DDI occurrences into 65 groups based on their descriptions. The dataset has 572 medicines, 74,528 interactions, and 65 event categories. The authors created a DDI multimodal deep learning (DDIMDL) framework to predict DDI instances by combining several pharmacological properties. Through 5-fold cross-validation, DDIMDL achieved an accuracy of 0.8852 and an area under the precision-recall curve of 0.9208 in computational trials, outperforming baseline techniques and other methods. Guggenmos et al. [33] developed a diagnostic ML classifier for alcohol dependency utilizing structural, functional task-based, and functional resting-state MRIs. They used a multimodal classification strategy on a massive dataset in-

cluding neuroimaging data from 119 alcoholic patients and 97 healthy controls. The multimodal method achieved a balanced accuracy of 79.3% by estimating optimal weightings for each modality and using training data to identify the best classifiers, slightly outperforming the best individual modality. Beattie et al. [34] identified major traits related to heroin use based on the National Survey on Drug Use and Health (NSDUH). Three RF methods were used to address the uneven use of heroin among respondents, which helped identify important features in nonlinear classification models. After stability testing, the best model's variable importance scores (VIS) showed the most important attributes for identifying heroin addiction. The results indicated that the optimal performance was reached by combining random oversampling with RF, as evidenced by an AUPRC of 0.5437. Kamarajan et al. [35] identified cognitive, personality, and impulsive characteristics that, when paired with the RF method, may distinguish healthy controls from adult males with AUD. The study examined 14 Default Mode Network (DMN) connections, two neuropsychological characteristics, and impulsive factors. When it came to differentiating between participants with AUD and those in good health, the RF model's classification accuracy was 76.67%.

III. MATERIAL AND METHODS

A. Proposed Methodology

The proposed methodology integrates a novel lightweight CNN with XAI to detect SDA , utilizing academic, social, and legal features while ensuring both predictive efficiency and interpretability. A dataset of 12,745 student records from universities and rehabilitation centers was collected, covering ten key attributes such as academic decline, social isolation, financial issues, and health problems. The dataset was pre-processed by cleaning, normalization, and transformation into structured arrays suitable for CNN models. Three baseline hybrid classifiers were designed: method-1 (1D CNNs-FLSs), method-2 (1D CNNs-LSTM-FLSs), and method-3, each extracting distinct latent representations. To enhance detection accuracy, we developed a weighted average ensemble (method-4) that combines the outputs of the three base methods through optimized weight assignment using Grid-Search. The ensemble detection is then refined using a softmax layer to classify students as addicted or non-addicted. To ensure interpretability, we employed SHAP (Shapley Additive Explanations) to analyze feature importance and explain model detections. This methodology not only provides high detection accuracy but also enhances transparency and trust in detecting SDA.

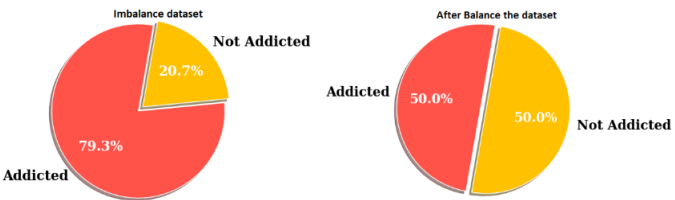


Fig. 1. Imbalance and balance data ratio in preprocessing stages.

B. Data preprocessing

In the preprocessing stage, data balancing was performed to mitigate class imbalance, where addicted instances (79.3%) largely outnumbered not addicted instances (20.7%). Such an imbalance can bias models toward the majority class, limiting their ability to recognize the minority class. By balancing the dataset to 50% per class, we ensured fair representation, reduced bias, and improved model generalization and reliability across all evaluation metrics. Fig. 1 presents the data ratio for the balance and imbalance datasets.

The datasets used in this study were publicly available and stored in CSV format. During preprocessing, the Pandas package was employed to read the dataset details, followed by cleaning procedures to remove null and redundant values in preparation for subsequent stages. This step also included label encoding, as deep neural networks require numerical inputs. Since the label column contained categorical values, we applied a One-Hot Encoder to transform them into numerical representations suitable for model training.

To solve this, we utilized two techniques: forward filling and imputation employing interpolation. Let q_x denote the value of a variable at time x . For the forward filling, the missing data q_x is assigned employing the most current available data, indicated as $q_x - 1$:

$$q_x = q_x - 1 \quad (1)$$

For missing data in a continuous variable, we use linear interpolation between the accessible data points. Provided two levels x_1 and x_2 , where the values q_{x1} and q_{x2} are familiar, the missing data for $x \in [x_1, x_2]$ are linearly interpolated as follows:

$$q_x = q_{x1} + \frac{x - x_1}{x_2 - x_1} \cdot (q_{x2} - q_{x1}) \quad (2)$$

This secures flat flows between values and neglects initiating biases because of sudden alters in the data.

Normalization was also applied during preprocessing to ensure that all feature values were within a comparable range. The raw data read from the CSV files exhibited variations in scale and distribution, which could negatively affect the learning performance of the model. To address this, we employed the StandardScaler from sklearn.preprocessing library, which standardizes the input features by shifting them to have zero mean and unit variance.

C. Origin method-1 (1D CNNs-FLSs)

This research utilizes 1D CNN [36] accompanied by fully linked systems (FLSs) to create the first standard method for the detection of SDA (SDA). 1D CNN executes adeptly with organized data. For tabular data, a 1D CNN brings out time knowledge inside the numeric features. The features from the CSV dataset are kept using an array to build a vector of attributes. This vector of attributes is provided for the proposed standard method as fed in. Using 7 subsequent units of feature layers (UFLs) that possess convolutional-1D (Conv-1D), max pooling-1D, batch normalization (BN), and dropout layers, the structure eliminates the secret domestic scene that

TABLE I
STRUCTURE OF THE DIFFERENT NUMBER OF UFL IN THE ORIGIN METHOD-1 (1D CNNs-FLSs).

UFL number	Layer Type	Kernel size	Number of kernels	Output size
UFL-1	Conv-1D	2	128	(2, 128)
	Maxpooling-1D	2	128	(2, 32)
	Dense	None	64	(None, 64)
UFL-2	Conv-1D	2	64	(2, 64)
	Dense	2	64	(2, 64)
UFL-3	Conv-1D	2	64	(2, 64)
	Maxpooling-1D	2	64	(2, 64)
	Dense	None	32	(None, 32)
UFL-4	Conv-1D	2	64	(2,64)
	Dense	2	64	(2,64)
UFL-5	Conv-1D	2	32	(2,32)
	Maxpooling-1D	2	32	(2,32)
	Dense	None	16	(None,16)

comes from the input data. Table I presents the layer-wise organization of the UFL block implemented in method-1 and method-2. It is displayed in Fig. 2. The 1D conv-1D, max pooling-1D, and BN are integral layers of UFLs. Two FLSs assemble the final universal features from the SDA data. The proposed 1D CNNs-FLSs method-1 grabs the preprocessed SDA data using the preprocessing techniques used in this study. Here, the initial UFL structure uses the kernel size of 2, padding='same', dilation rate=(1X1), and a stride of one. The amplification rate decreases the feed-in vector attributes. Then, the Rectified Linear Unit (ReLU) activates its result when BN is added. After resolving the disappearing gradient problem. The BN layer helps all of the layers in the neural network to learn in a normalized way. It helps to boost the learning procedure by normalizing secret layer triggers. To manage the problem of overfitting, we utilized the dropout layer and kernel regularization (L1 and L2), and the rate was 0.01. Then, we used the output of the first layer in the second layer. This is made up of 128 filters with a related kernel. Here, also works related stride and dilation rate. ReLU also helps to activate the output. Then, it's attached to the dropout rate of 0.30. BN is executed using the output being taken into a 1D max pooling, and also, the window size is 2. The next 4 UFLs use the filter of 64,64,64, and another one is 32 kernel size, dilation rate, and stalk organization was constant as the last structure. The flattening layer and the 30% dropout rate for the final UFL are also worked on. Then the result of the flattening surface is accepted by 2 FLSs of 64 and 32 units by utilizing the dropout rate of 20%. However, at last, the result layer uses softmax, which is used to detect SDA of the hidden attributes that are learned by utilizing UFLs. The UFL's structure is not fixed; it can be customized according to the task. The converts in the UFL organization initially returned in the convolution, dilation, pooling, and BN configurations.

D. Method-2 (1D CNNs-LSTM-FLSs)

Fig. 3 illustrates the architecture of method-2, which is derived from method-1 with an integrated modified LSTM. In this architecture, the ultimate UFL from the standard method-1 is retained, while a Universal Feature Acquisition Block

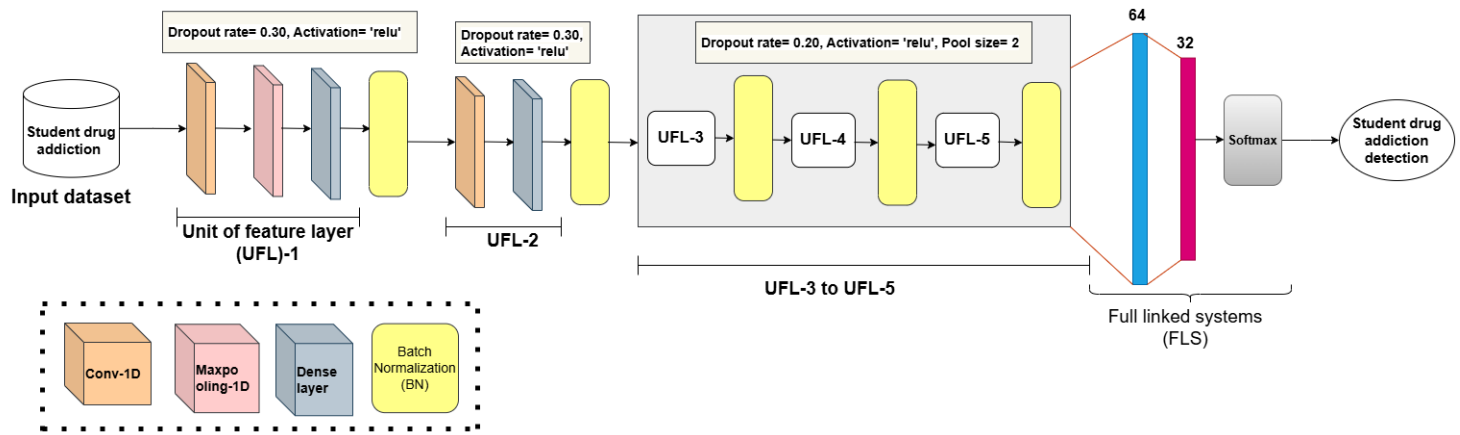


Fig. 2. Architecture of modified Origin method-1 (1D CNNs-FLSs).

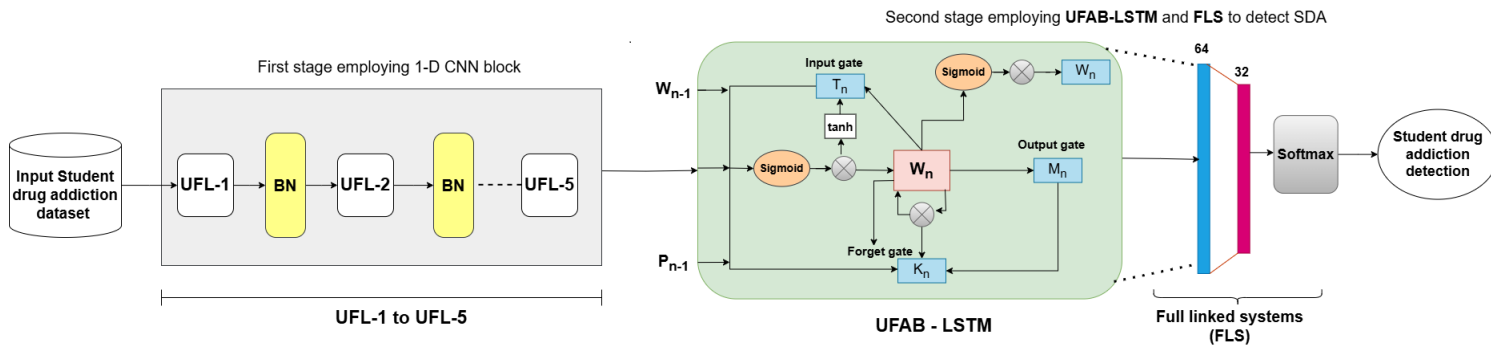


Fig. 3. Architecture of modified method-2 (1D CNNs-LSTM-FLSs).

(UFAB) incorporating an LSTM layer is introduced. The UFAB is designed to capture universal dependencies from the attributes generated by the UFLs and simultaneously regulate the universal weights. A 50% dropout layer is applied within the UFAB to reduce overfitting. The FLSS component remains unchanged from method-1. As shown in Fig. 3, the LSTM unit comprises three gates—forget, input, and output—that govern the flow of information into and out of the cell. These gates enable the method to preserve and update information across time steps, thereby ensuring effective long-term dependency learning. Here, the forget gate k_n denotes which vital info is to be kept back from the previous cell state W_n ; this is computed by Eq.(3). Here, the input gate T_n denotes what appropriate knowledge can be embraced from the ongoing time step. M_n presents the output gate, and it signifies the secret state, which would be forwarded to the next LSTM. T_n and M_n computed equations are displayed in Eq.(5) and Eq.(6). Initially, one sigmoid surface decides what service to bring up to date. Next, to control the whole system, there is a hyper tangent tanp surface layer that creates a feature vector W_n^o , and its probable values can be between -1 and 1. In the subsequent step, the information from the previous cell state is updated to the new cell state as represented in Eq. (7). The length of the attribute frame varies depending on

the number of features in the dataset. The LSTM effectively learns deep universal correlation attributes by maintaining a consistent length, achieved through selecting the outcome of the previous step from variable-length feature sets, prioritizing the most significant features. The output is then computed using Eq. (7), Eq. (8), and Eq. (9).

$$k_n = \omega(G_k * [p_{n-1}, z_n] + d_k) \quad (3)$$

Here, p_{n-1} is the secret later output at the last period step. z_n denotes the current step input, G_k is the weight matrix between T_n and M_n and d_k presents the connection bias at timestep n. ω denotes the logistic sigmoid function, which is computed using Eq.(4).

$$\omega(z) = \frac{1}{1 + \alpha^z} \quad (4)$$

$$T_n = \omega(G_x * [p_{n-1}, z_n] + d_x) \quad (5)$$

$$W_n^o = \tan p(G_y * [p_{n-1}, z_n] + d_y) \quad (6)$$

$$W_n = k_n * W_{n-1} + T_n * W_n M \quad (7)$$

$$M_n = \omega (G_1 * [p_{n-1}, z_n] + d_1) \quad (8)$$

$$p_n = M_n * \tan p(W_n) \quad (9)$$

Here, G_x and G_y are the weight matrices between M_n and T_n and ω respectively. b_y denotes the bias vector for G_y , W_n , W_{n-1} , k_1 , T_n denotes the info of cell state, last timestep, forget gate, and input gate at timestep n , respectively. The value created by $\tan p$ is W_n^o and where G_1 and d_1 denotes the weights and bias of the M_n at timestep n .

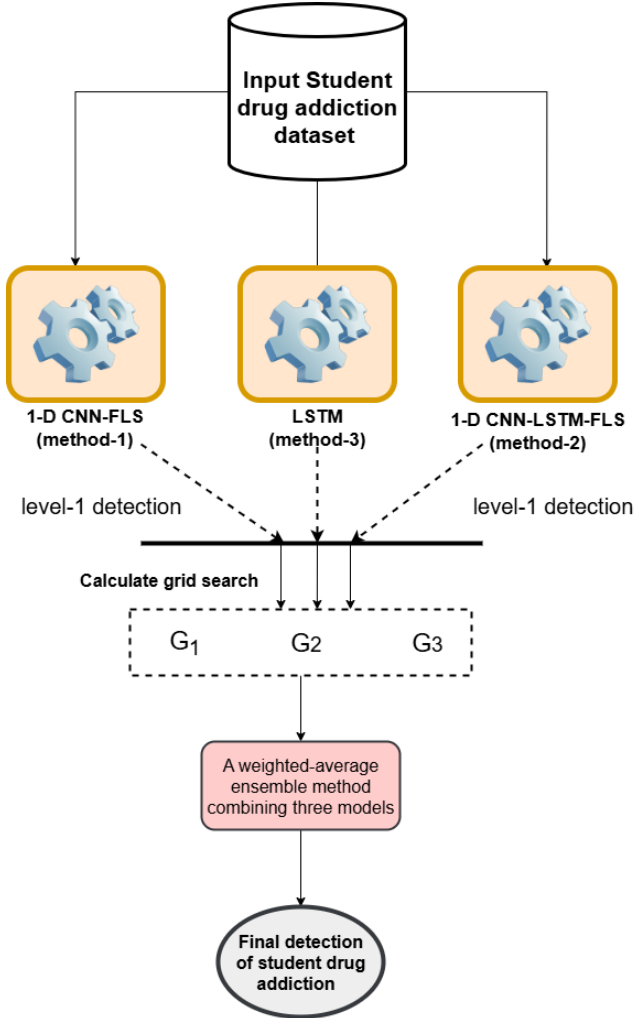


Fig. 4. Architecture of proposed ensemble method (method-4).

E. Proposed ensemble method-4

The learning processes of many methods are combined in ensemble learning (EL) [37] to produce a more accurate detection with a maximum level of stability and comprehensiveness than the accuracy of the individual ML methods. Certain models excel at simulating one aspect of the data, while others excel at simulating another. Because different models won't produce the same mistakes in the same test dataset, EL is successful. It guarantees the most trustworthy and accurate

detection. In this study, method-1, method-2, and method-3 were combined to create a weighted average ensemble (see Fig. 4). First, we used the Grid-Search technique to choose the best weights for each of the different methods. The class with the highest detected probability is then determined by multiplying the chosen optimal weights by the detection outcomes of each method, adding up the product of the components over the designated axis to determine the weighted detection result. The class with the highest anticipated probability is then selected for the final prediction based on this weighted prediction result. When tested using both the original and enhanced datasets, the ensemble method-4 outperformed the individual methods-1, 2, and 3 in terms of weighted average accuracy (WAA).

F. Shapley Additive Explanation

The Shapley value is the mean of the edge donation of all possible amalgamations. The Shapley merit is the only lucrative distribution mechanism that satisfies the below four parameters: effectiveness, similarity, linearity, and null player. By utilizing the idea of the Shapley merit, XAI introduces the result of the student- i : $\omega(y^{(i)})$ as the total of individual attributes- j 's addition $\beta_j(y_j^{(i)})$.

$$\beta_0 = \frac{1}{M} \sum_{i=1}^M \omega(y^{(i)}) \quad (10)$$

$$\beta_j(y_j^{(i)}) = \theta(y_j^{(i)}) - \frac{1}{M} \sum_{h=1}^M \theta(y_j^{(h)}) \quad (11)$$

$$\omega(y^{(i)}) = \beta_0 + \sum_{j=1}^H \beta_j(y_j^{(i)}) \quad (12)$$

Where M indicates the number of students. We deduced $K(\beta(Y_j)) = \frac{1}{M} \cdot \sum_{i=1}^M \beta_j(y_j^{(i)}) = 0$ from Eq. (11). The Shap merit has been justified to be congruous [38] and is available for every method, including the proposed ensemble method (ENM). The calculation time of naive SHAP computations optimizes exponentially with the number of attributes E . The connection between an attribute y_j and SHAP merit in ENM β_{ENM} is provided as follows [38].

Lundberg's main mechanism [39] utilizes the aggregation of the absolute value (L1-norm) of an individual student's SHAP merit to calculate the attribute's donation $\sum_{i=1}^M |\beta(y_j^{(i)})|$. However, we propose utilizing the variation (L2-norm) of the SHAP merit for calculating feature importance (FIM).

A SHAP summary plot [2] displays the attribute significance and an abstract of the SHAP reliance graphs. In the graph, attributes are ordered by their significance, as explained by Eq. (13-15), and collected vertically. Every row plot is an abstract of the SAHP reliance plot of the individual attribute Y_j .

$$FIM(Y_j) = Var(\beta(Y_j)) = \frac{1}{M} \cdot \sum_{i=1}^M [\beta(y_j^{(i)}) - K(\beta(Y_j))]^2 \quad (13)$$

TABLE II
DESCRIPTION OF THE DATASET

Features	Description	Values
Experimentation	Whether the student has experimented with drugs	(Yes/No)
Academic_Performance_Decline	Indicates if there's a decline in academic performance due to drug use	(Yes/No)
Social_Isolation	Whether drug use leads to social isolation	(Yes/No)
Financial_Issues	Indicates if the student faces financial difficulties due to drug use	(Yes/No)
Physical_Mental_Health_Problems	Presence of physical or mental health issues related to drug use	(Yes/No)
Legal_Consequences	Whether there are legal consequences associated with drug use	(Yes/No)
Relationship_Strain	Indicates if drug use strains relationships with others	(Yes/No)
Risk_Taking_Behavior	Engagement in risky behavior under the influence of drugs	(Yes/No)
Withdrawal_Symptoms	Presence of withdrawal symptoms when trying to quit drug use	(Yes/No)
Denial_and_Resistance_to_Treatment	Indicates denial or resistance to seeking treatment for drug addiction	(Yes/No)

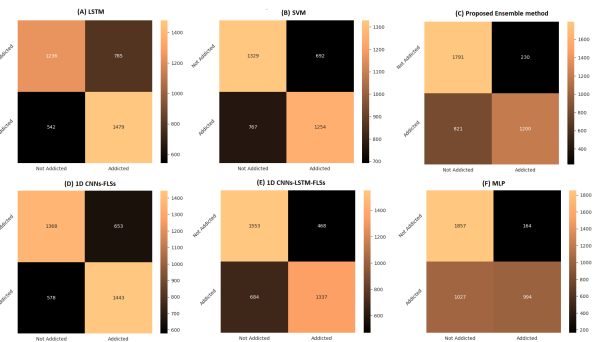


Fig. 5. Confusion matrices of the evaluated methods obtained with XAI.

$$= \frac{1}{M} \cdot \sum_{i=1}^M \left[\beta \left(y_j^{(i)} \right) \right]^2 - \{ K \left(\beta \left(Y_j \right) \right) \}^2 \quad (14)$$

$$= \frac{1}{M} \cdot \sum_{i=1}^M \left[\beta \left(y_j^{(i)} \right) \right]^2 \quad (15)$$

Employing our explanation, the variable significance of ENM in Eq. (16-17), in which all attributes are standardized, is provided as below:

$$FIM_{ENM} (Y_j) = \frac{1}{M} \cdot \sum_{i=1}^M \left[n_j y_j^{(i)} - n_j K (Y_j) \right]^2 \quad (16)$$

$$= \frac{n_j^2}{M} \cdot \sum_{i=1}^M \left[y_j^{(i)} - K (Y_j) \right]^2 \quad (17)$$

$$= |n_j|^2 \cdot Var (Y_j) = |\alpha_j|^2 \quad (18)$$

The ordering output is sorted by the absolute merit of the α_j , and is precisely the same as that of our explanation. Our explanation is also helpful for the attribute wrapping mechanism explained in the subsequent subsection.

IV. RESULTS

A. Dataset

This dataset provides a comprehensive exploration of factors related to drug addiction among university students. With features ranging from experimentation to denial of treatment, it offers valuable insights into the complex dynamics of addiction behaviors. The dataset includes a target variable classifying students as addicted or non-addicted based on these features. Utilize this dataset for ML tasks, predictive modeling, and behavioral analysis to better understand and address drug addiction issues among university populations. Table II presents the features and description of the dataset. This dataset contains ten important features related to drug addiction. We used 12745 student samples in this research. The dataset is publicly available at <https://www.kaggle.com/code/atifmasih/students-drugs-addiction-prediction/input>.

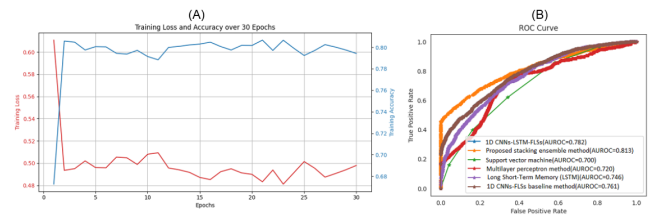


Fig. 6. (A) Training loss and accuracy for the proposed method (method-4) over 30 epochs, and (B) ROC curves for all methods after applying XAI.

B. Performance analysis

Fig. 5 presents the confusion matrices of models with XAI applied. The Proposed Ensemble Method and MLP achieved the lowest misclassification rates, with the highest true negatives (1791 and 1857). In contrast, LSTM and SVM showed higher false negatives (542 and 767), indicating lower efficient detection of addicted students. The hybrid 1D CNNs-LSTM-FLSs model exhibited a balanced sensitivity–specificity trade-off. Fig. 6 (A) shows the training loss and accuracy of the proposed method over 30 epochs, indicating rapid convergence and stable performance. Fig. 6 (B) presents ROC curves comparing the proposed stacking ensemble with AUROC of 0.813 against 1D CNN–LSTM–FLS with AUROC of 0.782 and 1D CNN–FLS with AUROC of 0.761, demonstrating the consistent superiority of the ensemble method. Fig. 7

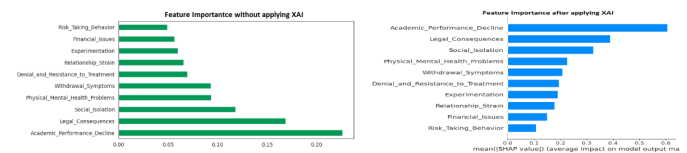


Fig. 7. Feature importance analysis without and with applying XAI.

illustrates the feature importance for both XAI and non-XAI approaches on our dataset. In both cases, features related to academic performance decline are the most influential for SDA. With XAI, these features achieved a SHAP value corresponding to 60% importance, compared to only 22% without XAI, demonstrating the effectiveness of our XAI approach for this dataset.

Fig. 8 demonstrates that academic performance decline, legal consequences, social isolation, and health problems exert the strongest global influence on the model outcome, with

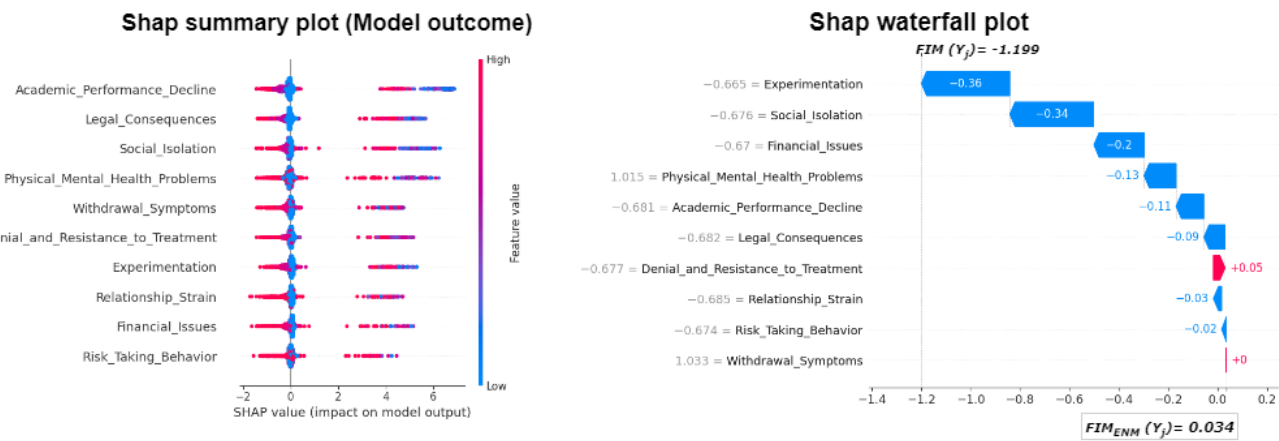


Fig. 8. SHAP analysis showing feature importance and individual prediction contributions, highlighting key factors such as academic decline, social isolation, and withdrawal symptoms.

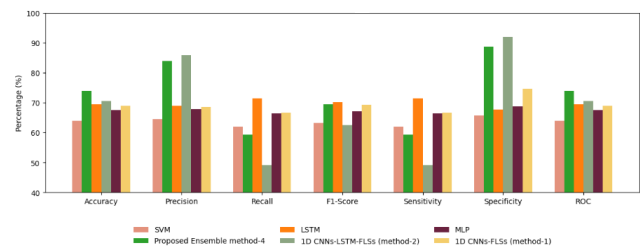
TABLE III
PERFORMANCE ANALYSIS FOR ALL EVALUATED METHODS USING XAI

Method	Accuracy%	Precision%	Sensitivity%	Specificity%	F1-score%	ROC%	Recall%	Log_Loss	Mathew_Corcoef%	Kappa%	MissClassification_Error%
SVM	63.90	64.44	62.05	65.76	63.22	63.90	62.05	13.01	27.83	27.81	29.44
Proposed Ensemble method- 4	74.00	83.92	59.38	88.62	69.55	74.00	59.38	9.30	50.19	48.00	26.00
LSTM (method-3)	69.54	68.85	71.40	67.69	70.10	69.54	71.40	10.98	39.12	39.09	30.46
1D CNNs-LSTM-FLSs (method- 2)	70.53	85.84	49.18	91.89	62.54	70.53	49.48	10.60	45.42	41.07	29.47
MLP	67.54	67.95	66.40	68.68	67.17	67.54	66.40	11.70	35.09	35.08	32.46
1D CNNs-FLSs (method-1)	68.98	68.66	66.65	74.57	69.23	68.98	69.82	11.18	37.96	37.95	29.44

SHAP values reaching approximately 6.0. At the individual detection level, negative contributions from experimentation -0.36 and social isolation -0.34 reduce the baseline output of -1.199 , while minor positive shifts from denial and resistance to treatment $+0.05$ and withdrawal symptoms $+0.00$ adjust the score upward, yielding a final detection of 0.034 . This underscores both the dominant global drivers of detection variability and the nuanced role of feature-level contributions in shaping individualized outcomes.

Table III showed that the proposed Ensemble method-4 outperforms the other tested methods across the majority of measures utilizing XAI. The method attains an accuracy of 74%, an F1-score of 69.55%, and a ROC of 74%. It also records the lowest log loss at 9.30% and a misclassification error of 26%. This indicates enhanced reliability and a reduction in overall errors. Additionally, the approach maintains an impressive balance between 83.92% precision and 88.62% specificity, making it more robust than methods such as LSTM and MLP, which have good sensitivity but are less reliable overall. While the method-2 demonstrates a high precision of 85.84% and specificity of 91.89%, its sensitivity is significantly lower at 49.18%, indicating challenges in accurately identifying positive cases. In contrast, Ensemble method-4 achieves a sensitivity of 59.38% while maintaining a strong specificity, enhancing its generalizability. Subsequently, the elevated Mathew's correlation coefficient of 50.19% and Kappa of 48% indicate a stronger alignment with the true outcomes. The results indicate that, with the support of XAI, the proposed ensemble method exhibits greater effectiveness and stability relative to alternative methods.

Performance comparison of different models across evaluation metrics using XAI-based analysis



Performance comparison of different models across evaluation metrics without XAI-based analysis

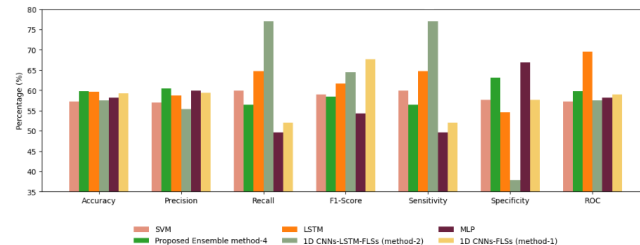


Fig. 9. Performance comparison of different models across multiple evaluation metrics with and without XAI-based analysis.

Fig. 9 shows that with XAI, the proposed ensemble achieves accuracy 75%, precision 85%, recall 86%, F1-score 70%, sensitivity 89%, specificity 91%, and ROC 72%. Without XAI, performance drops to around 60% across all metrics, confirming the effectiveness of XAI in enhancing detective accuracy and robustness.

Table IV presents a summary of the performance of the evaluated methods without the inclusion of XAI. The proposed

TABLE IV
PERFORMANCE ANALYSIS FOR ALL EVALUATED METHODS WITHOUT XAI

method	Accuracy%	Precision%	Sensitivity%	Specificity%	F1-score%	ROC%	Recall%	Log_Loss%	Mathew_Corrcoef%	Kappa%	MissClassification_Error%
SVM	57.27	56.92	59.85	57.69	58.97	57.27	59.85	15.40	14.54	14.54	42.73
Proposed Ensemble method- 4	59.80	60.50	56.44	63.16	58.40	59.80	56.44	14.59	19.65	19.59	40.20
LSTM (method-3)	59.64	58.76	64.69	54.59	61.58	69.54	59.64	14.55	19.38	19.28	40.36
1D CNNs-LSTM-FLSs (method- 2)	57.44	55.34	77.03	37.84	64.41	57.44	77.03	15.34	16.16	14.87	42.56
MLP	58.22	59.95	49.56	66.88	54.26	58.22	49.56	15.06	16.70	16.44	41.78
1D CNNs-FLSs (method-1)	59.23	59.35	52.08	57.64	67.64	58.97	59.23	58.61	14.70	18.46	40.14

Ensemble method-4 attains an accuracy of 59.80%, although its overall effectiveness is notably diminished in comparison to the XAI-supported scenario. The results represent a precision of 60.50% and a recall of 56.44%, yielding an F1-score of 58.40% and a ROC of 59.80%, which reflects a level of fair consistency. Additionally, the method’s log loss of 14.59% and misclassification error of 40.20% demonstrate that it is comparatively more reliable than a number of options in this particular context. Among the other methods, LSTM has the best recall and ROC (69.54% and 64.69%, respectively), but its balance is limited by its low specificity (54.59%). While method-2 obtains a respectable sensitivity and recall of 77.03%, the specificity falls well short at 37-84%, indicating a glaring reliance on false positives. In contrast, method-1 achieve the highest F1-score of 67.64% and the lowest misclassification error of 40.14%, despite having lower accuracy. Overall, the results show that method performance generally decreases in the absence of XAI, and that the suggested ensemble approach, although still competitive, gains notable stability and balancing benefits when XAI is included.

V. CONCLUSION AND FUTURE WORK

Millions of people in Bangladesh suffer from drug addiction, with the majority being young. Every day, more people become addicted to drugs. Therefore, drug abuse is becoming a greater issue. It not only negatively impacts our health but is also associated with unstable finances and, to a lesser degree, with terrorist activity. This research develops a novel ensemble framework for the early diagnosis of SDA that is strengthened with XAI. Using a large dataset of 12,745 student records, the methodology comprises academic, social, financial, health, and legal facts to show how ML models may provide accurate and individualized detections. The suggested framework adeptly addresses the intricate and multi-faceted character of SDA by including DL models, particularly 1D-CNN, LSTM networks, and FLS. The results demonstrate that when compared to conventional ML and DL approaches, the ensemble method that incorporates SHAP-based XAI performs substantially better. In particular, the suggested ensemble model demonstrates robustness and generalizability with an F1-score of 69.55%, a 74% accuracy rate, and great specificity and precision. With the addition of XAI, the model becomes much more interpretable, and important elements like academic degradation, social isolation, and withdrawal symptoms may be better understood.

The study relies on a dataset from rehabilitation institutions and universities, which could add bias and limit its generalizability, among other issues. Class imbalance may still impact

model performance, and important aspects like family history and peer pressure were not taken into account. The ensemble approach’s intricacy could also affect how well it works in the real world. In order to make the framework more generalizable and resilient, future work will center on improving its efficiency, adding more relevant factors, diversifying the dataset to include more different populations, and optimizing the model.

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